高级工程数学

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沈超敏

计算机科学与技术学院

cmshen@cs.ecnu.edu.cn

文附楼 105

今天内容

1. **上次内容**（Ch. 5、Ch. 6 (Topic 1. Introduction)）**的复习拓展**
2. Topic 2 Condition
3. Ch. 7
4. Gradient Methods
5. Newton's Methods

上次：Elements of Calculus

5.1 Sequences and Limits

5.2 Differentiability

5.3 The Derivative Matrix

5.4 Differentiation Rules

5.5 Level Sets and Gradients

5.6 Taylor Series

Topic1 的知识点、重点、难点

知识点

- 优化问题的一般形式：

$$\min_{\boldsymbol{x}\in\Omega} f(\boldsymbol{x})$$

  其中 $f(\boldsymbol{x})$ 为目标函数，Ω 为约束集合。
- 优化问题的主要分类：有约束 / 无约束、凸 / 非凸、光滑 / 非光滑。
- 局部最小点与全局最小点的概念。
- 梯度（Gradient）与 Hessian 矩阵在多维函数中的几何意义。

Topic 1 的知识点、重点、难点

重点

优化问题的核心在于通过一阶条件 $\nabla f(x) = 0$ 寻找候选极值点，并利用二阶条件（Hessian 的正定性）判断其为极小点；凸函数中局部最小即为全局最小，而含约束问题则常可通过变量消元化为无约束形式。

Topic 1 的难点

- 区分局部最小点、鞍点与约束最小点。
- 约束优化中如何判断最优解是否在边界上。

一句话总结:

本节核心在于理解优化问题的建模、分类与基本求解原则，掌握梯度和 Hessian 在判断极值中的作用，为后续最优条件的学习奠定基础。

上次：习题

6.1 Consider the problem

$$\begin{aligned} \text{minimize} \quad & f(\boldsymbol{x}) \\ \text{subject to} \quad & \boldsymbol{x} \in \Omega, \end{aligned}$$

where $f \in \mathcal{C}^2$. For each of the following specifications for Ω, $\boldsymbol{x}^*$, and f, determine if the given point $\boldsymbol{x}^*$ is: (i) definitely a local minimizer; (ii) definitely not a local minimizer; or (iii) possibly a local minimizer.

b. $f : \mathbb{R}^2 \to \mathbb{R}$, $\Omega = \{\boldsymbol{x} = [x_1, x_2]^\top : x_1 \geq 1, x_2 \geq 2\}$, $\boldsymbol{x}^* = [1, 2]^\top$, and gradient $\nabla f(\boldsymbol{x}^*) = [1, 0]^\top$.

答案：Possibly a local minimizer

习题

6.1 Consider the problem

$$\begin{aligned} &\text{minimize} && f(\boldsymbol{x}) \\ &\text{subject to} && \boldsymbol{x} \in \Omega, \end{aligned}$$

where $f \in \mathcal{C}^2$. For each of the following specifications for Ω, $\boldsymbol{x}^\star$, and f, determine if the given point $\boldsymbol{x}^\star$ is: (i) definitely a local minimizer; (ii) definitely not a local minimizer; or (iii) possibly a local minimizer.

c. $f : \mathbb{R}^2 \to \mathbb{R}$, $\Omega = \{\boldsymbol{x} = [x_1, x_2]^\top : x_1 \geq 0, x_2 \geq 0\}$, $\boldsymbol{x}^\star = [1, 2]^\top$, gradient $\nabla f(\boldsymbol{x}^\star) = [0, 0]^\top$, and Hessian $\boldsymbol{F}(\boldsymbol{x}^\star) = \boldsymbol{I}$ (identity matrix).

d. $f : \mathbb{R}^2 \to \mathbb{R}$, $\Omega = \{\boldsymbol{x} = [x_1, x_2]^\top : x_1 \geq 1, x_2 \geq 2\}$, $\boldsymbol{x}^\star = [1, 2]^\top$, gradient $\nabla f(\boldsymbol{x}^\star) = [1, 0]^\top$, and Hessian

$$\boldsymbol{F}(\boldsymbol{x}^\star) = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

C 是, D 不是

习题

6.1 Consider the problem

$$\begin{aligned} &\text{minimize} && f(\boldsymbol{x}) \\ &\text{subject to} && \boldsymbol{x} \in \Omega, \end{aligned}$$

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$$\boldsymbol{F}(\boldsymbol{x}^\star) = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

C 是, D 不是

今天: Topic 2: Condition

Three acronyms:

- FONC (First Order Necessary Condition)
- SONC (Second Order Necessary Condition)
- SOSC (Second Order Sufficient Condition)



interior pt $\nabla f(\boldsymbol{x}^*) = 0, \nabla^2 f(\boldsymbol{x}^*) \geqslant 0$

general pt $\nabla f(\boldsymbol{x}^*) \cdot \boldsymbol{d} \geqslant 0 \quad \forall d$

If $\nabla f(\boldsymbol{x}^*) \cdot \boldsymbol{d} = 0$ for some $\boldsymbol{d}, \boldsymbol{d}^\top \nabla^2 f(\boldsymbol{x}^*)\, \boldsymbol{d} \geqslant 0$

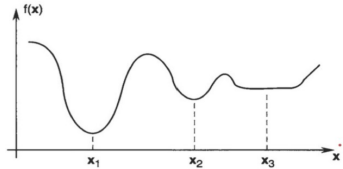
$\nabla f(\boldsymbol{x}^*) = 0, \nabla^2 f(\boldsymbol{x}^*) > 0$

Topic 2: Condition

★ Conditions for Local Minimizers

- Global minimizer $\boldsymbol{x}^*$: $f(\boldsymbol{x}) \geqslant f(\boldsymbol{x}^*) \quad \forall \boldsymbol{x} \in \Omega \backslash \{\boldsymbol{x}^*\}$
- Strict global minimizer $\boldsymbol{x}^*$: $f(\boldsymbol{x}) > f(\boldsymbol{x}^*) \quad \forall \boldsymbol{x} \in \Omega \backslash \{\boldsymbol{x}^*\}$
- Mathematically, "near" can be characterized as $\|\boldsymbol{x} - \boldsymbol{x}^*\| < \varepsilon$
- $\boldsymbol{x}^*$ is a local minimizer if $\exists \varepsilon > 0$, s.t.

$$f(\boldsymbol{x}) \geqslant f(\boldsymbol{x}^*) \quad \forall \boldsymbol{x} \in \Omega \backslash \{\boldsymbol{x}^*\} \,\&\, \|\boldsymbol{x} - \boldsymbol{x}^*\| < \varepsilon$$



Topic 2: Condition

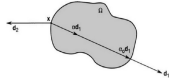
★ First Order Necessary Condition

Theorem. $\boldsymbol{x}^*$ is a local minimizer of f over Ω. Then for any feasible direction $\boldsymbol{d}$ at $\boldsymbol{x}^*$, we have

$$\nabla f\left(\boldsymbol{x}^*\right) \cdot \boldsymbol{d} \geqslant 0$$

Explanation. ① feasible direction $\boldsymbol{d}$ at a point $\boldsymbol{x} \in \Omega$ is a direction so that: starting from $\boldsymbol{x}$ and moving towards $\boldsymbol{d}$ remains in Ω.

Math language: $\exists \alpha_0 > 0$ s.t. $\boldsymbol{x} + \alpha \boldsymbol{d} \in \Omega, \forall \alpha \in [0, \alpha_0]$



② $\nabla f\left(\boldsymbol{x}^*\right) \cdot \boldsymbol{d}$: inner product of two vectors.

Also write as $\boldsymbol{d}^\top \nabla f\left(\boldsymbol{x}^*\right)$ or $\left(\nabla f\left(\boldsymbol{x}^*\right), \boldsymbol{d}\right)$, $\left\langle \nabla f(\boldsymbol{x}^*), \boldsymbol{d} \right\rangle$

$\frac{\partial f}{\partial \boldsymbol{d}} \triangleq \nabla f \cdot \boldsymbol{d}$ is the directional derivative when $\|\boldsymbol{d}\| = 1$

- ③ Define $\phi(\alpha) = f\left(\boldsymbol{x}^* + \alpha \boldsymbol{d}\right)$ for $\alpha \in [0, \alpha_0]$, then

$$\phi'(0) = \begin{cases} \lim_{\alpha \to 0^+} \frac{\phi(\alpha) - \phi(0)}{\alpha} = \lim_{\alpha \to 0^+} \frac{f(\boldsymbol{x}^* + \alpha \boldsymbol{d}) - f(\boldsymbol{x}^*)}{\alpha} & \text{Def} \\ \nabla f\left(\boldsymbol{x}^*\right) \cdot \boldsymbol{d} & \text{Chain rule} \end{cases}$$

Topic 2: Condition

- Proof. Let $\boldsymbol{d}$ be any feasible direction at $\boldsymbol{x}^*$.

  Define $\phi(\alpha) = f(\boldsymbol{x}^* + \alpha \boldsymbol{d})$, then

$$\begin{aligned} f(\boldsymbol{x}^* + \alpha \boldsymbol{d}) - f(\boldsymbol{x}^*) &= \phi(\alpha) - \phi(0) = \phi'(0)\alpha + o(\alpha) \\ &= \langle \nabla f(\boldsymbol{x}^*) \cdot \boldsymbol{d} \rangle \alpha + o(\alpha) \end{aligned}$$

  If $\boldsymbol{x}^*$ is a local minimizer,

  (i.e., $\exists \varepsilon$, s.t. $f(\boldsymbol{x}) \geqslant f(\boldsymbol{x}^*)$, $\forall \boldsymbol{x} \in \Omega \backslash \{\boldsymbol{x}^*\}$ & $\|\boldsymbol{x} - \boldsymbol{x}^*\| < \varepsilon$)

  for sufficiently small α (e.g. $\|\alpha \boldsymbol{d}\| < \varepsilon$), $f(\boldsymbol{x}^* + \alpha d) - f(\boldsymbol{x}^*) \geqslant 0$, then

$$\phi'(0) = \nabla f(\boldsymbol{x}^*) \cdot \boldsymbol{d} \geqslant 0$$

- FONC Two possibilities for a given feasible direction $\boldsymbol{d}$.

$$\begin{cases} \nabla f(\boldsymbol{x}^*) \cdot \boldsymbol{d} > 0 \text{ then } f(\boldsymbol{x}^* + \alpha \boldsymbol{d}) > f(\boldsymbol{x}^*) \text{ for all sufficienty small } \alpha > 0 \\ \nabla f(\boldsymbol{x}^*) \cdot \boldsymbol{d} = 0 \text{ . check second-order derivative} \end{cases}$$

Topic 2: Condition

★ Second Order Necessary Condition

- Theorem If $\boldsymbol{x}^*$ is a local minimizer of f over Ω, and there exists a feasible direction $\boldsymbol{d}$ at $\boldsymbol{x}^*$ s.t. $\nabla f\left(\boldsymbol{x}^*\right) \cdot \boldsymbol{d}=0$, then

$$\boldsymbol{d}^{\top} \nabla^2 f\left(\boldsymbol{x}^*\right) \boldsymbol{d} \geqslant 0$$

- Proof Consider $\phi(\alpha)=f\left(\boldsymbol{x}^*+\alpha \boldsymbol{d}\right)$ and its Taylor series at $\alpha=0$

$$\phi(\alpha)=\phi(0)+\phi'(0) \alpha+\frac{1}{2} \phi''(0) \alpha^2+o\left(\alpha^2\right).$$

$$\text{as } \phi'(0)=\nabla f\left(\boldsymbol{x}^*\right) \cdot \boldsymbol{d}=0$$

So we have $\phi(\alpha)-\phi(0)=\frac{1}{2} \phi''(0) \alpha^2+o\left(\alpha^2\right)$. Written in terms f is $f\left(\boldsymbol{x}^*+\alpha \boldsymbol{d}\right)-f\left(\boldsymbol{x}^*\right)=\frac{\alpha^2}{2} \boldsymbol{d}^{\top} \nabla^2 f\left(x^*\right) \boldsymbol{d}+o\left(\alpha^2\right)$.

If $\boldsymbol{d}^{\top} \nabla^2 f\left(\boldsymbol{x}^*\right) \boldsymbol{d}<0$, then for sufficiently small α (how small?)

which contradicts that x^* is a local minimizer.

so, $\boldsymbol{d}^{\top} \nabla^2 f\left(\boldsymbol{x}^*\right) \boldsymbol{d} \geq 0$.

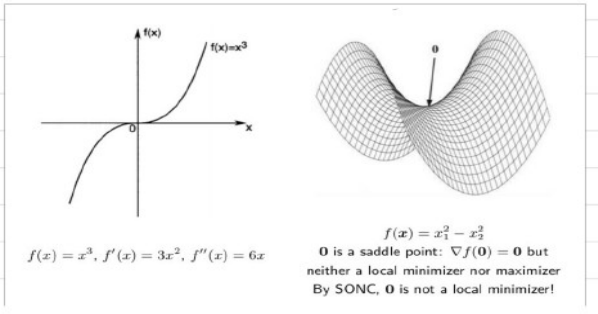
Topic 2: Condition

Corollary $\boldsymbol{x}^*$ is an interior local minimizer of f. Then

- FONC $\nabla f(\boldsymbol{x}^*) = 0$
- SONC $\boldsymbol{d}^\top \nabla^2 f(\boldsymbol{x}^*)\, \boldsymbol{d} \geqslant 0, \quad \forall \boldsymbol{d} \in \mathbb{R}^n$

  Examples. 6.3 (p.86), 6.5 (p.89)

  Necessary conditions are not sufficient



$f(x) = x^3,\ f'(x) = 3x^2,\ f''(x) = 6x$

$f(\boldsymbol{x}) = x_1^2 - x_2^2$
$\mathbf{0}$ is a saddle point: $\nabla f(\mathbf{0}) = \mathbf{0}$ but neither a local minimizer nor maximizer
By SONC, $\mathbf{0}$ is not a local minimizer!

Topic 2: Condition

★ Second Order Sufficient Condition

- Th 6.3 (SOSC) $f \in C^2(\Omega), \boldsymbol{x}^* \in \Omega$ is an interior point.

  Suppose that (1) $\nabla f(\boldsymbol{x}^*) = \mathbf{0}$; (2) $\nabla^2 f(\boldsymbol{x}^*) \succ 0$.

  Then $\boldsymbol{x}^*$ is a strict local minimizer of f.

  Pf. $\nabla^2 f(x^*) \succ 0 \Leftrightarrow \lambda_{\min}\left(\nabla^2 f(x^*)\right) > 0$

  (Prove by diagonaligation of $\nabla^2 f(\boldsymbol{x}^*) = Q^\top \wedge Q\Big)$

  For a feasible direction $\boldsymbol{d} \neq 0$, define $\phi(\alpha) = f(\boldsymbol{x}^* + \alpha \boldsymbol{d})$

$$f(\boldsymbol{x}^* + \alpha \boldsymbol{d}) - f(\boldsymbol{x}^*) = \phi(\alpha) - \phi(0) = \frac{1}{2}\phi''(0)\alpha^2 + o\left(\alpha^2\right)$$

$$= \frac{1}{2}\boldsymbol{d}^\top \nabla^2 f(\boldsymbol{x}^*)\, \boldsymbol{d}\alpha^2 + o\left(\alpha^2\right)$$

$$\geqslant \frac{1}{2}\lambda_{\min}\|d\|^2\alpha^2 + o\left(\alpha^2\right) > 0$$

  if α is sufficiently small.

Topic 2: Condition



interior pt $\nabla f(\boldsymbol{x}^*) = 0, \nabla^2 f(\boldsymbol{x}^*) \geqslant 0$

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习题

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一定不是

习题

6.1 Consider the problem

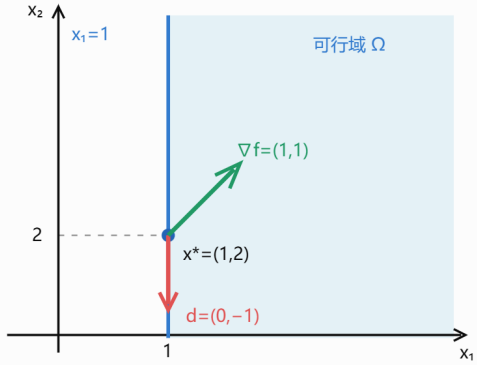
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一定不是

3. 几何直观



- 绿色箭头：梯度方向，函数值上升最快。
- 红色箭头：可行的下降方向，函数值减小。
- 蓝色直线：约束边界 $x_1 = 1$。

Ch 7. One-Dimensional Search Methods

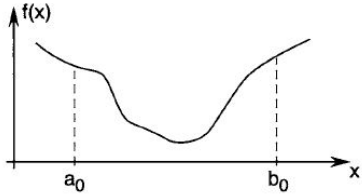


Figure 7.1 Unimodal function.

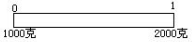
7.1 Introduction
7.2 Golden Section Search
7.3 Fibonacci Method
7.4 Bisection Method
7.5 Newton's Method
7.6 Secant Method
7.7 Bracketing
7.8 Line Search in Multidimensional Optimization

Ch 7. One-Dimensional Search Methods

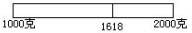
§2 单 因 素

我们知道，钢要用某种化学元素来加强其强度，太少不好，太多也不好.例如，碳太多了成为生铁，碳太少了成为熟铁，都不成钢材，每吨要加多少碳才能达到强度最高？假定已从理论上算出)每吨在1000克到2000克之间.普通的方法是加1001克，1002克，……，做下去，做了一千次以后，才能发现最好的选择，这种方法称为均分法.做一千次实验既浪费时间又浪费原材料.为了迅速找出最优方案，我们建议以下的“折迭纸条法”.

请牢记一个数0.618.



用一个有刻度的纸条表达1000～2000克，在这纸条长度的0.618的地方划一条线，在这条线所指示的刻度做一次实验，也就是按1618克做一次实验.



然后把纸条对中迭起，前一线落在另一层上的地方，再划一条线，这条线在1382克处，再按1382克做一次实验.



两次实验进行比较，如果1382克的好一些，我们在1618处把纸条的右边一段剪掉，得：



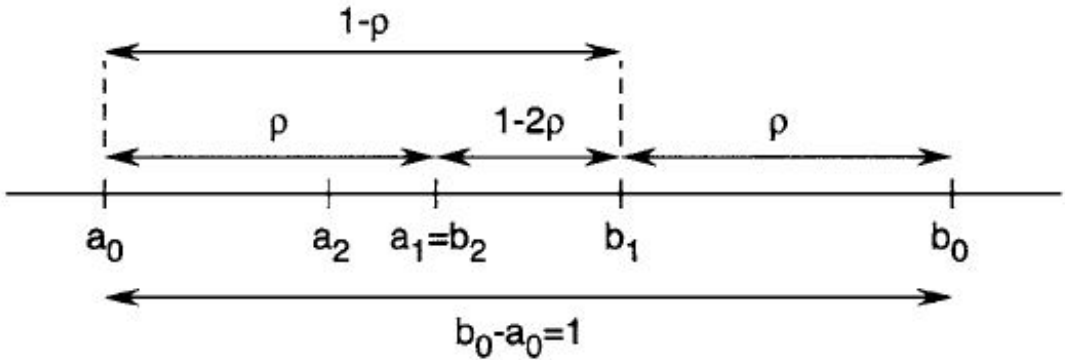
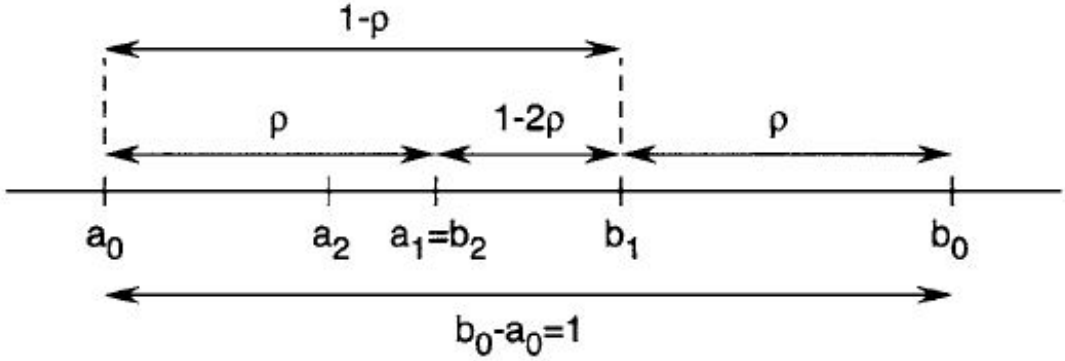


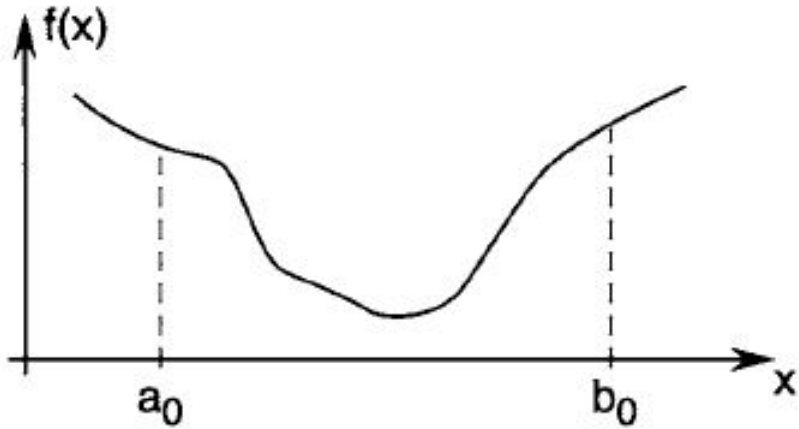
Figure 7.4 Finding value of ρ resulting in only one new evaluation of

7.3 Fibonacci Method



Ch 7. One-Dimensional Search Methods

7.4 Bisection Method



Topic 3: Ch 8. Gradient Methods

$\nabla f(x)$ is the direction of maximum rate of increase of f at x.
$-\nabla f(x)$ is the direction of maximum rate of decrease of f at x.

Lemma: (Cauchy-Schwarz inequality) For $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{R}^n$

$$\boldsymbol{a} \cdot \boldsymbol{b} = (\boldsymbol{a}, \boldsymbol{b}) = \sum_{i=1}^{n} a_i b_i \leqslant \left(\sum_{i=1}^{n} a_i^2 \right)^{\frac{1}{2}} \left(\sum_{i=1}^{n} b_i^2 \right)^{\frac{1}{2}} = \|\boldsymbol{a}\| \cdot \|\boldsymbol{b}\|$$

Pf: Consider

$$\begin{aligned} y(t) &= \left(\sum a_i^2 \right) t^2 + \left(2 \sum a_i b_i \right) t + \sum b_i^2 \\ &= \sum_{i=1}^{n} \left(a_i^2 t^2 + 2 a_i b_i t + b_i^2 \right) \\ &= \sum \left(a_i t + b_i \right)^2 \geqslant 0 \end{aligned}$$

$y(t) = 0$ iff $a_i t + b_i = 0, \forall i$

So $\Delta = \left(2 \Sigma a_i b_i \right)^2 - 4 \left(\Sigma a_i^2 \right) \left(\Sigma b_i^2 \right) \leqslant 0$ □

Topic 3: Gradient Methods

Apply to $(\nabla f, \boldsymbol{d}) \leqslant \|\nabla f\| \cdot \|\boldsymbol{d}\| = \|\nabla f\|$ if $\|\boldsymbol{d}\| = 1$

Equality holds $\Leftrightarrow \boldsymbol{d} = \frac{\nabla f}{\|\nabla f\|}$

Therefore $-\nabla f(\boldsymbol{x})$ is the max-rate descending direction. When $\nabla f(\boldsymbol{x}) \neq \boldsymbol{0}$, for α sufficiently small, $f(\boldsymbol{x} - \alpha \nabla f(\boldsymbol{x})) < f(\boldsymbol{x})$.

Topic 3: Gradient Methods

Gradient Descent Algorithm:

Start from $\boldsymbol{x}^0$ for $k = 0, 1, 2, \cdots$ till converge

$$\boldsymbol{x}^{k+1} = \boldsymbol{x}^k - \alpha_k \nabla f\left(\boldsymbol{x}^k\right)$$

Ideal condition: $\left\|\nabla f\left(\boldsymbol{x}^k\right)\right\| = 0$

Practical conditions:

gradient condition $\left\|\nabla f\left(\boldsymbol{x}^k\right)\right\| < \varepsilon$

success objective condition $\frac{\left|f\left(\boldsymbol{x}^{k+1}\right) - f\left(\boldsymbol{x}^k\right)\right|}{\left|f\left(\boldsymbol{x}^k\right)\right|} < \varepsilon$

successive point difference $\frac{\left\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\right\|}{\left\|\boldsymbol{x}^k\right\|} < \varepsilon$

Replace denominator by $\max\left\{1, \left|f\left(\boldsymbol{x}^k\right)\right|\right\}$ or $\max\{1, \|\boldsymbol{x}^k\|\}$ to avoid division by tiny numbers.

Topic 3: Gradient Methods

Step size:

1. Exact line search. Expensive and not worth

2. Fixed estimate value.

3. Line search (Ch. 7).

具体地

1. Exact or "best" for $\phi_k(\alpha) = f\left(x_k - \alpha \nabla f\left(x_k\right)\right)$

$$\alpha_k = \underset{\alpha \geqslant 0}{\arg\min}\, \phi_k(\alpha)$$

This is called Steepest Descent method.

梯度下降法的性质：相邻方向垂直

$$\alpha_k = \arg\min_{\alpha \geq 0} f(x_k - \alpha \nabla f(x_k)).$$

令

$$\phi(\alpha) = f(x_k - \alpha \nabla f(x_k)).$$

由于 α_k 是内部极小点，所以

$$\phi'(\alpha_k) = 0.$$

利用链式法则，

$$\phi'(\alpha) = -\nabla f(x_k - \alpha \nabla f(x_k))^T \nabla f(x_k).$$

代入 α_k，得到

$$-\nabla f(x_{k+1})^T \nabla f(x_k) = 0.$$

因此

$$\boxed{\nabla f(x_{k+1})^T \nabla f(x_k) = 0}$$

Topic 3: Gradient Methods

2. Based on properties of f, choose a fixed value

small: converges slow

large: may diverge faster

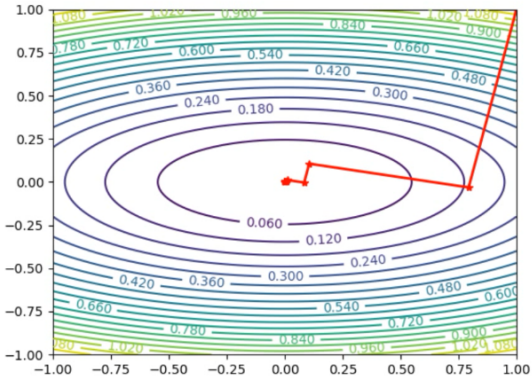
convergence efficiency

3. Several practical line search algorithms

• Golden section • Newton's method

• Fibonacci • Secant method

梯度下降法缺点



we need $t = O(1/\epsilon)$ iterations to have $\|\nabla f(x^k)\| \leq \epsilon$.

作业

编写一个完整的最速下降算法，

1. 定义 $f(\boldsymbol{x})$

2. 给出 $\boldsymbol{x}^*$ 的算法

运行时，输入 $f(\boldsymbol{x})$ 和 $\boldsymbol{x}_0$

输出 $\boldsymbol{x}^*$

Topic 4: Newton's method

Consider $\min_{x \in \mathbb{R}^n} f(x)$

- $n = 1$ $\quad x_{k+1} = x_k - \left(f''(x_k)\right)^{-1} f'(x_k)$
- $n > 1$ $\quad x_{k+1} = x_k - \left(\nabla^2 f(x_k)\right)^{-1} \nabla f(x_k)$

Another form ① solve $\nabla^2 f(x_k)\, d_k = -\nabla f(x_k)$

② update $x_{k+1} = x_k + d_k$

Remark. Do not require $\nabla^2 f(x_k) > 0$ only needs non-singular (invertible).

Namely, Newton's method also works for non-convex optimization problems.

but may not find local min.

Topic 4: Newton's method

Pro. 1. Convergences super-fast (quadratic rate).

2. Affine invariant.

Con. 1. Local convergence. Require $\|x_0 - x^*\|$ is small enough.

2. Computational cost.

Form Hessian matrix $O(n^2)$. Compute $(\nabla^2 f)^{-1}$: $O(n^3)$.

Topic 4: Newton's method

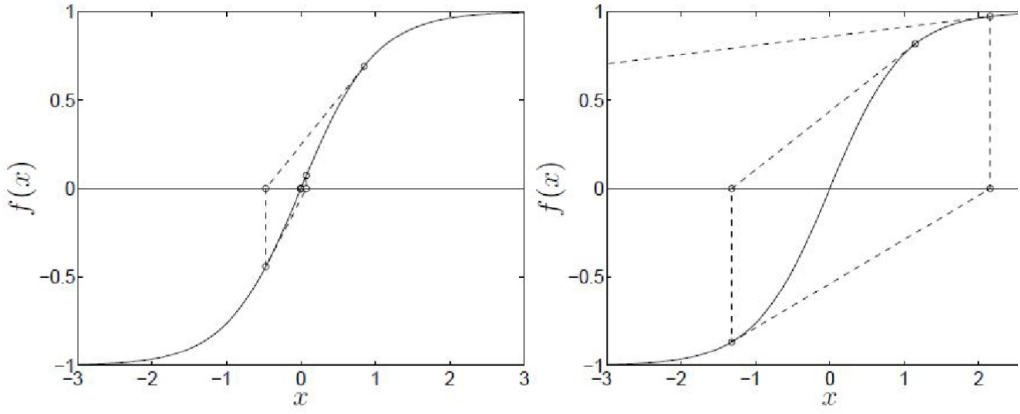


Figure 12.3 The solid line in the left plot is $f(x) = (e^x - e^{-x})/(e^x + e^{-x})$. The dashed line and the circles indicate the iterates in Newton's method for solving $f(x) = 0$, starting at $x^{(0)} = 0.85$ (left) and $x^{(0)} = 1.15$ (right). In

Topic 4: Newton's method

Derivation. Given current approximate x_k, approximates f by its quadratic Taylor series

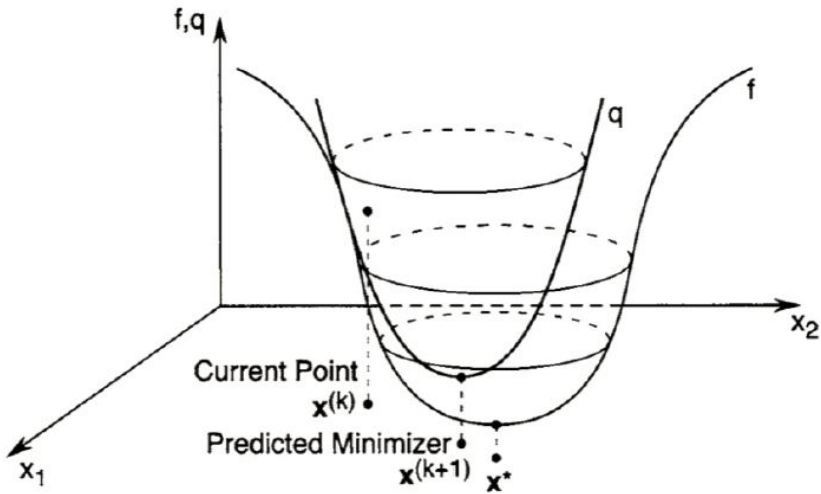
$$f(x) \approx f_q(x; x_k) := f(x_k) + (\nabla f(x_k),\ x - x_k) + \frac{1}{2}(\nabla^2 f(x_k)(x - x_k),\ x - x_k)$$

$$\min_{x \in \mathbb{R}^n} f(x) \rightsquigarrow \min_{x \in \mathbb{R}^n} f_q(x; x_k) \rightsquigarrow \nabla f_q(x_{k+1}; x_k) = 0.$$

$$\nabla f_q(x; x_k) = \nabla f(x_k) + \nabla^2 f(x_k)(x - x_k)$$

$\downarrow$ Newton's method

Topic 4: Newton's method



Convergence Analysis.

Theorem. Suppose $f \in C^3$. x^* is a critical pt, i.e. $\nabla f(x^*) = 0$, and $\nabla^2 f(x^*)$ is invertible. Then for all x_0 sufficiently close to x^*, Newton's method is well defined for all k, and $\|x_{k+1} - x^*\| \leq C \|x_k - x^*\|^2 \quad \forall\, k = 0, 1, 2, \cdots$

Proof. Denote by $F(x) = \nabla^2 f(x)$. Then $\det F(x) \in C^1$. As $\det F(x^*) \neq 0$, for sufficiently small ε, $\det F(x) \neq 0$, $\forall \|x - x^*\| < \varepsilon$. So $F(x)$ is invertible.

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Furthermore $\|F^{-1}(x)\| \leq C$, $\forall \|x - x^*\| < \varepsilon$.

Assume x_k satisfies $\|x_k - x^*\| < \varepsilon$, then $F^{-1}(x_k)$ exists and $\|F^{-1}(x_k)\| \leq C$.

Then $$x_{k+1} - x^* = x_k - x^* - (\nabla^2 f(x_k))^{-1} \nabla f(x_k)$$
$$= (\nabla^2 f(x_k))^{-1} \left[\nabla^2 f(x_k)(x_k - x^*) - \nabla f(x_k) \right].$$

We apply first order Taylor expansion to $\nabla f(x^*)$ at x_k to get
$$\nabla f(x^*) = \nabla f(x_k) + \nabla^2 f(x_k)(x^* - x_k) + O(\|x_k - x^*\|^2)$$

Note that $\nabla f(x^*) = 0$ and the sign change, we have
$$\nabla^2 f(x_k)(x_k - x^*) - \nabla f(x_k) = O(\|x_k - x^*\|^2).$$

Therefore $$\|x_{k+1} - x^*\| \leq C \|(\nabla^2 f(x_k))^{-1}\| \|x_k - x^*\|^2$$
$$\leq C_1 \|x_k - x^*\|^2$$

Again by choosing ε sufficiently small s.t. $C_1\varepsilon^2 < \varepsilon$, we conclude

$\|x_{k+1} - x^*\| < \varepsilon$ and $F(x_{k+1})^{-1}$ exists and $\|F(x_{k+1})^{-1}\| \leq C$.

So if ε is small enough and $\|x_0 - x^*\| < \varepsilon$, all $\|x_k - x^*\| < \varepsilon$ and

$$\|x_{k+1} - x^*\| \leq C_1 \|x_k - x^*\|^2 \quad \forall\ k = 0, 1, 2, \cdots$$

which implies the local quadratic convergence. #.

Modification of Newton's method.

Newton's method may not be a descent method, i.e. $f(x_{k+1}) > f(x_k)$ is possible (e.g. x^* is a local maximum). Have to restrict to stricktly convex functions.

Lemma. Assume $\nabla^2 f(x) > 0$, $\forall x$. If $\nabla f(x_k) \neq 0$, then Newton's direction $d_k = -(\nabla^2 f(x_k))^{-1} \nabla f(x_k)$ is a descent direction in the sense that $f(x_k + \alpha d_k) < f(x_k)$ for sufficiently small α.

Proof. Let $\phi(\alpha) = f(x_k + \alpha d_k)$. Then $\phi'(\alpha) = (\nabla f(x_k + \alpha d_k), d_k)$ and

$\phi'(0) = -(\nabla f(x_k), (\nabla^2 f(x_k))^{-1} \nabla f(x_k)) = -(g_k, g_k)\alpha < 0$ where

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For convex functions, we can use the following modification

1. Compute d_k by solving $\nabla^2 f(x_k)\, d_k = -\nabla f(x_k)$.
2. Find $\alpha_k = \operatorname{argmin} f(x_k + \alpha d_k)$ by line search.
3. Update $x_{k+1} = x_k + \alpha_k d_k$.

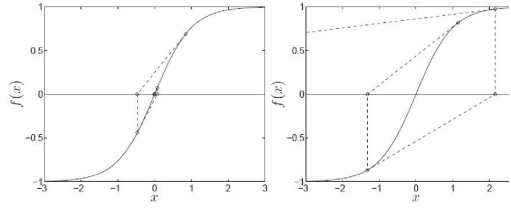


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What if $\nabla^2 f$ is not SPD? Note that for non-convex functions, the gradient method $x_{k+1} = x_k - \alpha_k I \nabla f(x_k)$ is always a descent method. This motivates the Levenberg-Marquardt modification

$$x_{k+1} = x_k - \alpha_k \left(\nabla^2 f(x_k) + \mu_k I \right)^{-1} \nabla f(x_k),$$

where $\mu_k > 0$ is chosen s.t. $\nabla^2 f(x_k) + \mu_k I > 0$ and $\alpha_k > 0$ is a step size.

It is a mixture of Newton and gradient methods:

- $\mu_k = 0$. Newton's method.
- $\mu_k \to +\infty$. Gradient method.